Liangwei Hu

Robotics Systems & Embodied AI | MuJoCo / LeRobot | ROS2 Simulation-to-Real

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M.S. Candidate, USTC | Class of 2027
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PROFILE

M.S. candidate at the University of Science and Technology of China with hands-on experience in **robot learning, physics simulation, ROS2-based real robots, and hardware–software integration**. Built end-to-end LeRobot/PyTorch workflows for PushT and LIBERO using ACT, SAC, and SmolVLA, including a custom MuJoCo environment and contact-dynamics debugging. Also developed a ROS2 mobile-robot workflow spanning Gazebo simulation and physical deployment with vision, sensing, communication, and motor control.

Robot Learning	Simulation & Physics	Sim ↔ Real Systems
LeRobot / PyTorch / ACT / SAC / SmolVLA	MuJoCo / Gazebo / LIBERO / contact tuning	ROS2 / LiDAR / depth camera / MCU & motors

EDUCATION

University of Science and Technology of China (USTC) | M.S. in Quantum Science and Technology	Sep 2024 — Jun 2027
Research: SPAD detector/readout systems; current focus expanded to robot learning, embodied AI, and simulation systems.	
Wuhan University | B.S. in Physics	Sep 2020 — Jun 2024
GPA: 3.86/4.00 | Overall rank: 4 | Coursework: Analog/Digital Electronics, Circuits, Microcontrollers, Quantum Mechanics	

TECHNICAL SKILLS

Robot Learning	PyTorch, LeRobot, ACT, SAC, SmolVLA; imitation learning, reinforcement learning, VLA closed-loop inference and evaluation
Simulation	MuJoCo, Gazebo, LIBERO, Gym-style environments; contact-dynamics tuning, environment alignment, simulation evaluation
Robotics	ROS/ROS2, Raspberry Pi, LiDAR, depth camera, upper/lower-controller communication, sensor/actuator integration, system debugging
Software	Python, C, Linux, Git; training/inference pipelines, dataset/model management, metrics and video visualization
Hardware	Verilog, FPGA test logic, PCB design, microcontrollers, motor control, board-level bring-up and debugging

PROJECT EXPERIENCE

LeRobot × MuJoCo Robot Learning & Embodied Simulation Workspace
Aug 2026 — Present

Personal Project | PushT + LIBERO | ACT / SAC / SmolVLA | GitHub: HUliangwei/robot

LeRobot · PyTorch · MuJoCo · ACT · SAC · SmolVLA · LIBERO

- **Built an end-to-end robot-learning workflow** for PushT and LIBERO covering datasets, training, checkpoints, closed-loop inference, evaluation metrics, and video results in a reproducible LeRobot workspace.
- **Implemented a custom MuJoCo PushT environment** aligned with the official gym-pusht observation/action/reward/success semantics, enabling the same ACT policy to run closed-loop in both pymunk and MuJoCo.
- **Debugged simulation fidelity at the contact-dynamics level:** traced excessive post-push sliding/rotation to MuJoCo soft-contact behavior, tuned damping/friction to recover stop-after-push behavior, and improved rollout reward sum by ~104%.
- **Trained and validated policies:** self-trained ACT for 25k steps (L1≈0.12), brought up a SAC learner–actor RL pipeline, and reproduced community ACT performance up to 0.9534 coverage in the official PushT environment and 0.865 in MuJoCo.
- **Validated a VLA manipulation pipeline** on Franka + MuJoCo in LIBERO; evaluated official SmolVLA on LIBERO-Spatial task0 with 4/5 successful episodes (80%), exercising image + state + language instruction → action closed-loop inference.

ROS2 Mobile Robot — Simulation ↔ Real-System Validation
Undergraduate

Gazebo | Python + YOLO | ROS2 | Raspberry Pi | MCU Motor Control | Provincial Electronic Design Competition, 2nd Prize

- Built a Gazebo-based mobile-robot and sensor workflow to validate motion, perception, and task logic, then deployed the corresponding ROS2 interfaces on a physical robot using a Raspberry Pi, LiDAR, depth camera, and low-level motor controller.
- Developed the Python/YOLO upper-computer pipeline for target/symbol recognition and task logic; ROS2 communicated commands to the lower microcontroller, which executed motor drive and vehicle motion, completing a perception–decision–communication–execution loop.
- Mapped real-system sensor/control interfaces and failure cases back into simulation for reproducible debugging and regression checks, forming a bidirectional simulation–real engineering workflow rather than a simulation-only demo.

SPAD Sensing & FPGA Readout System
Sep 2024 — Present

Graduate Research | Sensor Hardware, FPGA, and System Engineering

- Designed the SPAD board-level readout, FPGA test board, PCB, and Verilog test logic; owned timing generation, signal acquisition, interface definition, and system bring-up.
- Extended the board-level research prototype into a 1×16-channel dedicated readout IC, gaining hands-on experience in complex sensing systems, hardware interfaces, root-cause debugging, and engineering integration.